

Changyu (Thomason) Zhao

608-886-7470 | czhao249@wisc.edu | [linkedin.com/in/changyu-zhao](https://www.linkedin.com/in/changyu-zhao) | github.com/ThomasonZhao

EDUCATION

University of Wisconsin - Madison

Bachelor of Science in Computer Science (Honors) and Math

Cumulative GPA: **3.9/4.0**, Honor: **Dean's List** 2021 - 2023

Sep. 2021 – May 2025

Madison, Wisconsin

WORK EXPERIENCE

Tencent Xuanwu Lab

Security Researcher Intern

Aug. 2023

Beijing, China

- Developed a **hardware-assisted** dynamic control flow analysis tool for **Linux & Windows** platform based on Last Branch Record (LBR), an Intel hardware feature that records the branch jump addresses.
- Implemented **kernel modules/drivers** to enable LBR and collect branch records from the kernel.
- Designed a **user-mode** analysis library to reconstruct the control flow graph from the collected branch records.

UW - Madison Cybersecurity Operation Center

Cybersecurity Student Intern Analyst

Feb. 2023 – Present

Madison, Wisconsin

- Analyzed daily outbound malicious traffic reports from Palo Alto Firewall and Cisco Secure Endpoint (AMP).
- Traced malicious traffic in **Elastic** and **Palo Alto Panorama** to notify security alert to related user or department.
- Conducted **security authorization** and **Identity and Access Management (IAM)** for access restrictive accounts.
- Engaged in regular communication with team members and users, fostering a **collaborative environment** to address security concerns and mitigate potential threats.

RESEARCH EXPERIENCE

ASU SEFCOM/CDF Security Research Apprenticeship

Research Assistant

May. 2023 – Aug. 2023

Tempe, Arizona

- Developed and implemented a security research tool named "**LibHopper**," a **Python** based automated **exploitation primitives finding framework** for arbitrary user libraries.
- Leveraged automated processes to streamline the evaluation of security implications related to **memory corruption vulnerabilities** in library internal structures, enhancing the capabilities of security researchers and developers.
- Employed a multifaceted methodology involving custom **GDB (GNU Debugger) helper scripts** and **symbolic execution** techniques to conduct comprehensive exploitability analysis.
- Identified and cataloged **assets of exploitable primitives** within library internal states, derived from meticulous analysis of memory corruption scenarios and symbolic execution findings.

Tencent Spark Early Career Program

Research Assistant | Remote

Jun. 2021 – Present

Shenzhen, China

- Conducted source code review of the modified Vijos Online Judge (OJ) system, identifying deserialization and sandbox escape vulnerabilities.
- Exploited the identified vulnerabilities, gaining root privileged reverse shell within Docker container.
- Demonstrated privilege escalation to root level on host machine by leveraging CVE-2019-5736 to escape from privileged Docker container.
- Utilized IDA for reverse engineering router firmware, located buffer overflow vulnerability mentioned in CVE-2018-8879.
- Applied **Return Oriented Programming (ROP)** techniques to establish reverse shell and achieve full control over the router.

PROJECTS

- Tenda AC18 AC1900 Router Exploitation** | *IDA, GDB, QEMU, Python* May 2022 – Aug.2022
- Exploited unfixed Common Vulnerabilities and Exposures (CVE) in Tenda AC18 router firmware.
 - Employed qemu-system to emulate arm32 router firmware on an x86-64 machine, enabling systematic debugging.
 - Developed automated stack overflow and command injection detection plug-in for IDA at decompiled code level.
 - Achieved Remote Code Execution (RCE) by exploiting vulnerabilities identified by the IDA plug-in.
 - Discovered multiple unfixed CVEs: CVE-2018-16334, CVE-2018-5767, CVE-2018-5768, CVE-2018-5770, CVE-2020-15916, CVE-2022-28556, CVE-2022-28557.
- Windows Calculator Reverse Engineering** | *IDA, Windbg, Frida* Jan. 2022 – Apr. 2022
- Exploited the Win10 and Win32 Calculator to manipulate arithmetic operations, resulting in intentionally incorrect results (e.g., $1 + 1 = 3$).
 - Conducted both static and dynamic analysis on the calculator binary using IDA and Windbg, unraveling its inner workings.
 - Utilized Frida to apply hooks on the calculator's library functions, enabling the hijacking of arbitrary arithmetic operations.

EXTRACURRICULAR ACTIVITIES

- r3kapiG CTF Team** Sep. 2021 – Present
Team member
- Achieved a notable accomplishment by securing a Rank 12/1282 worldwide in the DEF CON CTF Finals.
 - Participated in over 20 48-hour Capture the Flag (CTF) competitions since 2021, with a focus on Binary Exploitation (PWN) and Reverse Engineering challenges.
- pwn.college** Dec. 2021 – Jul. 2022
Independent Learning *Tempe, Arizona*
- Attained the Yellow Belt and Blue Belt levels by completing all 472 challenges in the Dojo instance, currently ranked 43 out of 2500+ participants.
 - Acquired comprehensive skills in various areas of binary exploitation, including program interaction, shellcoding, privilege escalation, reverse engineering, stack corruption, heap corruption, race conditions, and kernel security.

TECHNICAL SKILLS

Programming Languages: C/C++, Python, Java, x86-Assembly JavaScript, HTML/CSS, $\text{\LaTeX}$
Binary Security Tools: IDA/Ghidra, GDB/Windbg, Angr, Frida, Elastic Search, Joern
Web Security Tools: Burpsuit, Wireshark, Metasploit Framework, Kali Linux